25900067 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCECC102 - FUNDAMENTALS OF OS AND LINUX

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **1** marks

1. Name the technological innovation that led to the development of time-sharing systems. [K] / [CO1]
2. How the status of the current activity of a process can be identified? [U] / [CO2]
3. A system allocates memory in fixed-size blocks, allowing processes to be stored in non-contiguous locations. What is this memory management technique called? [K] / [CO3]
4. What way an Operating System helps the computer? [U] / [CO1]
5. Priority Scheduling lead to starvation.Why? [U] / [CO2]
6. Two processes require 200MB each. The system has free blocks of 150MB, 250MB, and 300MB. One process is allocated in the 250MB block. Name the fragmentation? [U] / [CO3]
7. Which OS structure stores information about a process? [K] / [CO2]
8. Why is a segment table needed in segmentation? [U] / [CO3]
9. Yes or No: A process moves directly from the 'new' state to the 'running' state. [K] / [CO2]

10. How does the system interpret an object file? [U] / [CO3]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Identify the different system calls for device management . [U] / [CO1]

12. What are the differences between CPU-bound and I/O-bound processes? [K] / [CO2]

13. How does the acyclic graph directory improve file sharing compared to a tree-structured directory? [U] / [CO3]

14. Explain how processes move between different scheduling queues. [U] / [CO2]

15. Explain why linked allocation is better suited for files that frequently grow in size. [U] / [CO3]

16. Identify the operating system services for the system. [U] / [CO1]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Identify the different types of operating systems. [U] / [CO1]

18. Explain the concept of a Resource Allocation Graph in operating systems.Provide an example to illustrate the detection of deadlock using Resource Allocation Graph. [U] / [CO2]

19. What is the purpose of a page replacement algorithm?List the different types of page replacement algorithms. [K] / [CO3]

[10x2 = 20]